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Ruei-Che Chang

Education

2022–Present **University of Michigan**, *Ann Arbor, Michigan*
Ph.D. Student in Computer Science & Engineering.
Human-AI Lab, advised by Anhong Guo.

2020–2022 **National Taiwan University**, *Taipei, Taiwan*
Visiting Student and Research Assistant.
Interactive Graphics Lab, advised by Robin Bing-Yu Chen.

2019–2021 **Dartmouth College**, *Hanover, New Hampshire*
M.S. in Computer Science.

2014–2018 **National Cheng Kung University**, *Tainan, Taiwan*
B.S. in Electrical Engineering.

Awards

2022 Rackham Travel Grant Award for UIST 2022.

2022 University of Michigan CSE Departmental First-Year Fellowship 2022.

2020 Best Paper Honorable Mention for Glissade [C.2] at ACM CHI 2020. [Top 5% of 3126 submissions]

2019 Dartmouth College 75% Tuition Scholarship.

Peer-Reviewed Conference Papers

2023 [C.9] **Ruei-Che Chang***, Seraphina Yong*, Fang-Ying Liao, Chih-An Tsao, Bing-Yu Chen. "Understanding (Non-)Visual Needs of the Design of Laser Cut Architecture." (*To Appear at* **CHI'23**). Hamberg, Germany. 2023.

2022 [C.8] **Ruei-Che Chang**, Chao-Hsien Ting, Chia-Sheng Hung, Wan-Chen Lee, Liang-Jin Chen, Yu-Tzu Chao, Bing-Yu Chen, Anhong Guo. "OmniScribe: Authoring Immersive Audio Descriptions for 360° Videos." *In The 35th Annual ACM Symposium on User Interface Software and Technology* (**UIST'22**). Bend, Oregon. 2022. [Acceptance Rate: 26.3%]

[C.7] Ching-Wen Hung, **Ruei-Che Chang**, Hong-Sheng Chen, Chung-Han Liang, Liwei Chan, Bing-Yu Chen. "Puppeteer: Exploring Intuitive Hand Gestures and Upper-Body Postures for Manipulating Human Avatar Actions." *In The 28th Annual ACM Symposium on Virtual Reality Software and Technology* (**VRST'22**). Tsukuba, Japan. 2022. [Acceptance Rate: 26.7%]

2021 [C.6] **Ruei-Che Chang**, Chih-An Tsao, Fang-Ying Liao, Seraphina Yong, Tom Yeh, Bing-Yu Chen. "Daedalus in the Dark: Designing for Non-Visual Accessible Construction of Laser-Cut Architecture." *In The 34th Annual ACM Symposium on User Interface Software and Technology* (**UIST'21**). Virtual Event. 2021. [Acceptance Rate: 21%]

[C.5] **Ruei-Che Chang***, Wen-Ping Wang*, Chi-Huan Chiang, Te-Yen Wu, Zheer Xu, Justin Luo, Bing-Yu Chen, Xing-Dong Yang. "AccessibleCircuits: Adaptive Add-On Circuit Components for People with Blindness or Low Vision." *In Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (**CHI'21**). Virtual Event, Japan. 2021. [Acceptance Rate: 26.3%]

2020 [C.4] **Ruei-Che Chang***, Chi-Huan Chiang*, Shuo-wen Hsu, Chih-Yun Yang, Da-Yuan Huang, Bing-Yu Chen. 2020. "TanGo: Exploring Expressive Tangible Interactions on Head-Mounted Displays." *In Symposium on Spatial User Interaction* (**SUI'20**). Virtual Event. 2020. [Acceptance Rate: 31%]

[C.3] **Ruei-Che Chang***, Yi-Shyuan Chiang*, Yi-Lin Chuang, Shih-Ya Chou, Hao-Ping Lee, I-Ju Lin, Jian Hua Jiang Chen, Yung-Ju Chang. "Exploring the Design Space of User-System Communication for Smart home Routine Assistants." *In Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (**CHI'20**). Virtual Event. 2020. [Acceptance Rate: 24.3%]

[C.2] Kai-Chieh Huang, Chen-Kuo Sun, Da-Yuan Huang, Yu-Chun Chen, **Ruei-Che Chang**, Shuo-wen Hsu, Chih-Yun Yang, Bing-Yu Chen. "Glissade: Generating Balance Shifting Feedback to Facilitate Auxiliary Digital Pen Input." *In Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems.* (**CHI'20**). Virtual Event. 2020. [Acceptance Rate: 24.3%] **[Best Paper Honorable Mention, Top 5% of 3126 submissions]**

2019 [C.1] Chi Wang, Da-Yuan Huang, Shuo-Wen Hsu, Chu-En Hou, Yeu-Luen Chiu, **Ruei-Che Chang**, Jo-Yu Lo, Bing-Yu Chen. "Masque: Exploring Lateral Skin Stretch Feedback on the Face with Head-Mounted Displays." *In Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology* (**UIST'19**). New Orleans, LA. 2019. [Acceptance Rate: 24.4%]

Posters, Demos, Workshops, Works in Progress

2022 [A.1] Ching-Wen Hung, **Ruei-Che Chang**, Hong-Sheng Chen, Chung-Han Liang, Liwei Chan, Bing-Yu Chen. "Puppeteer: Manipulating Human Avatar Actions with Intuitive Hand Gestures and Upper Body Postures." *In The 35th Annual ACM Symposium on User Interface Software and Technology* (**UIST Poster 2022**). Bend, Oregon. 2022.

Research Experience

Sep 2022 – present **Human-AI Lab, University of Michigan**
Graduate Research Assistant, advised by Anhong Guo
- Developed accessible XR experience using sound manipulations [work in progress].

Sep 2021 – Apr 2022 ***Remote research intern, advised by Anhong Guo***
Developed OmniScribe for authoring immersive audio descriptions for 360° videos [C.8].

Jun 2020 – Jul 2022 **Interactive Graphics Lab, National Taiwan University**
Research Assistant, advised by Bing-Yu Chen and Tom Yeh (Univ. of Colorado)
- Developed Daedalus for non-visual accessible construction of laser-cut architecture [C.6].
- Conducted study to understand (non-)visual needs for laser-cut model design [C.9].

Feb 2019 – Aug 2019 ***Research Assistant, advised by Bing-Yu Chen and Da-Yuan Huang (Huawei)***
Developed Unity applications for Glissade[C.2] and Masque [C.1].
- Developed TanGo for expressive haptic interaction on VR headset [C.4].

Sep 2019 – Sep 2020 **XDiscovery Lab, Dartmouth College**
Graduate student, advised by Xing-Dong Yang
- Developed AccessibleCircuits, a set of 3D-printed add-ons to existing circuit components [C.5].

Sep 2018 – Apr 2019 **Mobile and Ubiquitous Interaction Lab, National Yang Ming Chiao Tung University**
Research Assistant, advised by Yung-Ju (Stanley) Chang
- Designed and conducted an experiment to understand the communication between human and smart home agent [C.3].

Work Experience

2018 **Mandatory Military Service**, Taiwan.

2018 **Undergraduate research intern** at *Industrial Technology Research Institute (ITRI)*, Hsinchu, Taiwan.

Academic Services

Programm Committee: CHI '23 LBW

Reviewer: CHI('22 '23), UIST('22 '23), CSCW('23), TEI('23), SUI('23), DIS('22), ISS('22), MobileHCI('22), IEEE VR('23), CHI LBW('20 '21 '22)

Student Volunteer: UIST'22

Student Mentoring

2022–present **Minyu Cai**, *Undergrad student* at University of Michigan. (ImageExplorer deployment)

2022–present **Dier Hou**, *Undergrad student* at University of Michigan. (ImageExplorer deployment)

2022–present	**Chia-Sheng Hung**, *Master student* at National Taiwan University.
2022–present	**Wan-Chen Lee**, *Master student* at National Taiwan University. (Project TacNote)
2021–2022	**Fang-Ying Liao**, *Master student* at National Taiwan University.
2021–2022	**Chao-Hsien Ting**, *Master student* at National Taiwan University. (OmniScribe [C.7], UIST'22)
Jan–Apr 2021	**Chih-An Tsao**, *Master student* at National Taiwan University. (Daedalus [C.6], UIST'21)

Skills

Programming	Java, C#, Python, JavaScript, ROS, OpenCV, MongoDB, Swift, LaTeX
Prototyping	Arduino, 3D-printing, Fusion 360, Laser-cutting
Platforms/IDE	Unity3D, Android Studio, Fusion 360, Xcode

References

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